Variants of the GARCH Model & Regime Switching

Week 2 – Problem Set (Day 3)

FI 362 – Advanced Financial Econometrics | Shanghai Jiao Tong University

Instructions. Attempt every problem before turning to the solutions – the derivation matters more than the final number. Notation follows the lectures: $a_t = \sigma_t \varepsilon_t$ with $\varepsilon_t \overset{iid}{\sim} (0,1)$, $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$ is the conditional variance, and a GARCH$(1,1)$ is $\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2$. The quantity $\alpha + \beta$ is the *persistence*. These problems accompany the Day-3 lecture on IGARCH, GARCH-M, EGARCH and Markov switching. **Full step-by-step solutions are given at the end** – consult them only after attempting each part. All empirical figures are reproducible from `Day3_variants_code.R` and `Day3_regime_code.R` on the course `data/` folder.

1. GARCH(1,1): persistence at the edge

(25 pts)

A daily equity return series (in %, zero mean) is modelled as

$$\sigma_t^2 = 0.02 + 0.09\, a_{t-1}^2 + 0.90\, \sigma_{t-1}^2 \qquad \text{(units: } \%^2\text{)}.$$

(a) Compute the persistence $\alpha+\beta$ and the volatility *half-life* $H = \ln(0.5)/\ln(\alpha+\beta)$ (in trading days). Interpret.

(b) Compute the long-run (unconditional) daily volatility.

(c) With Gaussian innovations, a *finite unconditional kurtosis* exists iff $(\alpha+\beta)^2 + 2\alpha^2 < 1$. Check the condition, and if it holds evaluate

$$\kappa = 3\, \frac{1 - (\alpha+\beta)^2}{1 - (\alpha+\beta)^2 - 2\alpha^2}.$$

Comment on the size of κ.

(d) A shock puts the one-step-ahead variance at four times its long-run level. Using $\sigma_t^2(h) - \sigma^2 = (\alpha+\beta)^{h-1}\big(\sigma_t^2(1) - \sigma^2\big)$, find how many days h pass until the forecast is within 5% of the long-run level. *Guess before you compute – the answer is surprising.*

2. IGARCH: RiskMetrics and the flat forecast

(20 pts)

The RiskMetrics exponentially-weighted moving average (EWMA) of squared returns is

$$\sigma_t^2 = (1-\lambda)\, a_{t-1}^2 + \lambda\, \sigma_{t-1}^2, \qquad \lambda = 0.94.$$

(a) Write this as a GARCH$(1,1)$: identify ω, α, β and compute $\alpha+\beta$. Why is the model called *integrated* (IGARCH)?

(b) Prove that the h-step-ahead variance forecast equals the one-step forecast for *every* horizon h (a flat term structure of volatility).

(c) The EWMA places weight $(1-\lambda)\lambda^{j-1}$ on the squared return j days ago. What fraction of the total weight lies on the most recent 20 trading days? (Use $\sum_{j=1}^{20}(1-\lambda)\lambda^{j-1} = 1-\lambda^{20}$.)

(d) In one or two sentences: why is a flat volatility term structure both *convenient* for a risk desk and *dangerous* after a crisis?

3. GARCH-in-mean: the price of risk

(20 pts)

A GARCH-M model puts conditional volatility into the mean equation,

$$r_t = \mu + \delta\,\sigma_t + a_t, \qquad a_t = \sigma_t \varepsilon_t,$$

and estimation on daily S&P 500 returns gives $\hat{\mu} = -0.02$ and $\hat{\delta} = 0.11$ (daily %, with σ_t in %/day).

(a) Compute the expected daily return on a calm day ($\sigma_t = 1\%$) and on a turbulent day ($\sigma_t = 3\%$).

(b) Take the average daily risk premium $\hat{\delta} \cdot \mathbb{E}[\sigma_t]$ with $\mathbb{E}[\sigma_t] \approx 1.2\%$/day, and annualise it ($\times 252$). Is the resulting number economically plausible? What does that tell you about daily GARCH-M estimates?

(c) Give two reasons why $\hat{\delta}$ is frequently *statistically insignificant* when estimated on daily data.

4. EGARCH: asymmetry, quantified

(20 pts)

An EGARCH(1, 1) (Nelson) models the log-variance,

$$\log \sigma_t^2 = \omega + \alpha\, z_{t-1} + \gamma\big(|z_{t-1}| - \mathbb{E}|z|\big) + \beta\, \log \sigma_{t-1}^2, \qquad z_{t-1} = \frac{a_{t-1}}{\sigma_{t-1}},$$

with $\alpha = -0.16$, $\gamma = 0.14$, $\beta = 0.98$ and Gaussian innovations, so $\mathbb{E}|z| = \sqrt{2/\pi} = 0.798$. Let $g(z) = \alpha z + \gamma\big(|z| - \mathbb{E}|z|\big)$ be the "news impact" on the log-variance (holding σ_{t-1} fixed).

(a) Evaluate $g(z)$ at a one-day loss $z = -2$ and at an equal gain $z = +2$.

(b) By what *factor* does the $z = -2$ shock scale the variance relative to the $z = +2$ shock? (Compare $e^{g(-2)}$ with $e^{g(+2)}$.)

(c) Which sign of α produces the leverage effect, and what does $\beta \approx 0.98$ imply about the memory of the log-variance? What practical constraint does the log formulation remove relative to plain GARCH?

5. Regime switching (Markov switching)

(15 pts)

A two-state Markov-switching model fitted to daily S&P 500 returns gives a *calm* regime $\mathcal{N}(0.08,\, 0.70^2)$ and a *storm* regime $\mathcal{N}(-0.11,\, 1.95^2)$ (means and standard deviations in %/day), with transition probabilities $p_{11} = 0.99$ (stay calm) and $p_{22} = 0.97$ (stay in storm).

(a) Compute the expected duration of each regime, $1/(1 - p_{ii})$ days.

(b) Compute the stationary (long-run) probability of each regime, $\pi_1 = \dfrac{1 - p_{22}}{2 - p_{11} - p_{22}}$, $\pi_2 = 1 - \pi_1$.

(c) Compute the unconditional variance of the return as a two-component mixture, $\mathrm{Var}(r_t) = \sum_s \pi_s(\sigma_s^2 + \mu_s^2) - \big(\sum_s \pi_s \mu_s\big)^2$.

(d) Explain, using the Lamoureux–Lastrapes (1990) argument, why fitting a *single-regime* GARCH to data generated by this process would report a persistence $\hat{\alpha} + \hat{\beta}$ close to 1.